Artificial Intelligence-Driven Malware Detection via Windows Event Log Engineering in Smart Grid Control Systems: A Preliminary Study

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Abstract—Smart grid Industrial Control Systems (ICS) are high-value targets for advanced persistent threats. Windows Event Logs generated by SCADA hosts provide a rich, native telemetry surface for detecting malicious activity; however, the volume, heterogeneity, and sequential nature of event sequences make manual analysis intractable. This paper presents a preliminary study of an AI-driven detection framework that (i) synthesises realistic Windows Event Log sequences for smart grid environments—covering lateral movement, persistence, privilege escalation, and reconnaissance—(ii) engineers session-level feature vectors capturing event frequency, temporal dynamics, host cardinality, rare-event indicators, and Shannon sequence entropy, and (iii) classifies sessions using a Random Forest with stratified cross-validation. Experiments on a synthetic dataset of 6,000 labelled sessions demonstrate a macro F1-score of 1.00 on held-out data, establishing a strong baseline for future evaluation on operational logs. The proposed framework introduces measurable success criteria aligned with NERC CIP and IEC 62351 compliance objectives.

Index Terms—Windows Event Log, SCADA security, intrusion detection, smart grid, random forest, feature engineering, ICS malware

I. Introduction

The electric power grid has undergone a fundamental transformation over the last two decades. The integration of digital communications, programmable logic controllers, and IP-connected SCADA and Human-Machine Interface (HMI) systems has enabled unprecedented operational efficiency but has simultaneously expanded the cyber attack surface. High-profile incidents—most notably the Stuxnet worm [1] and the Ukrainian power grid attacks of 2015 and 2016—demonstrate that adversaries are willing and able to traverse the IT/OT boundary and compromise energy infrastructure.

Regulatory frameworks respond to this threat. The North American Electric Reliability Corporation Critical Infrastructure Protection (NERC CIP) standards mandate event logging, monitoring, and incident response for Bulk Electric Systems. IEC 62351 specifies security for power systems communications, including requirements for audit trail generation. Together they establish Windows Event Logs—produced natively by every Windows host in a smart grid operations centre—as a mandated, machine-readable record of system behaviour.

Despite this regulatory mandate, Windows Event Logs have received comparatively little attention as a detection surface for ICS-specific attacks. The research community has focused on network-level anomalies (e.g., Modbus/TCP deviations [2]) or dedicated ICS protocols, leaving host-based event logs underexplored. Meanwhile, general system-log anomaly detection has advanced significantly through deep learning approaches such as DeepLog [3] and LogAnomaly [4], and through log-parsing methods like Drain [5]. These techniques have not been applied in the ICS/SCADA context.

This paper makes the following contributions:

- A synthetic Windows Event Log generator that models four ICS-relevant attack patterns within a realistic smart grid topology.
- A session-level feature engineering pipeline tailored to the temporal structure of Windows Event Logs.
- A baseline evaluation of Random Forest classification under stratified cross-validation, establishing measurable success criteria.
- An open-source framework enabling reproducible ICS security research without access to operational data.

Section II reviews related work. Section III states the central hypothesis and research objectives. Section IV describes the methodology. Section V reports preliminary results, and Section VI concludes.

II. Related Work

A. Windows Event Log Anomaly Detection

Windows Event Logs contain semantically rich, structured records of system activity. He et al. [5] introduced Drain, an online log-parsing tree that maps raw log messages to templates in $O(\log n)$ time; their approach is directly applicable to tokenising Windows Event Log descriptions at scale. Du et al. [3] modelled log key sequences as a workflow using LSTM networks, detecting anomalies as deviations from learned sequential patterns. Their CCS 2017 evaluation demonstrated low false-positive rates

on the HDFS log dataset. Zhao et al. [4] extended this to LogAnomaly, capturing both sequential and quantitative anomalies via template-based semantic vectors. Garcia-Teodoro et al. [6] provided a foundational survey of anomaly-based network intrusion detection, whose classification taxonomy—statistical, knowledge-based, and machine-learning—applies equally to host-based log analysis. Collectively, these works demonstrate the viability of sequential log modelling but leave the ICS-specific event-identifier semantics unexploited.

B. ICS/SCADA Intrusion Detection

Industrial control system security demands domain-adapted detection strategies. Carcano et al. [7] proposed a multidimensional critical-state analysis that flags SCADA command sequences deviating from learned safe states, achieving 100% detection on a simulated water treatment testbed. Goldenberg and Wool [2] built deterministic finite automata from Modbus/TCP network traffic, detecting protocol violations with zero false positives in a power utility environment. Mitchell and Chen [8] surveyed IDS techniques for cyber-physical systems, categorising approaches by detection method (misuse vs. anomaly) and system layer (network, process, device). Goh et al. [9] released the Secure Water Treatment (SWaT) dataset, enabling reproducible evaluation of process-level anomaly detectors. Zhu et al. [10] formalised a taxonomy of cyber attacks on SCADA systems across all Purdue model levels, which informs the attack patterns modelled in this work. A key gap identified across this body of work is the absence of Windows host-log analysis as a detection layer within the ICS hierarchy.

C. AI/ML Methods for Intrusion Detection

Machine learning for intrusion detection has a long history. Liao et al. [11] reviewed supervised, unsupervised, and hybrid methods, noting that ensemble methods consistently outperform single classifiers on tabular security data. Breiman's Random Forest [12]—a bagged ensemble of decision trees—remains a strong baseline due to its resistance to overfitting, built-in feature importance ranking, and native support for class imbalance through per-class sample weighting. Chen and Guestrin's XGBoost [13] extended gradient boosting to large-scale tabular problems, achieving state-of-the-art results on the KDD Cup 1999 network intrusion dataset. Tavallaee et al. [14] revealed severe redundancy in the KDD Cup 1999 benchmark, motivating the community towards domain-specific synthetic datasets. Sommer and Paxson [15] argued that the closed-world assumption of supervised ML makes deployment in open operational environments inherently difficult, underscoring the need for realistic synthetic training data and rigorous evaluation protocols. NIST SP 800-82 [16] provides the governing reference for ICS security controls, including detection and response requirements that bound the target performance metrics for this work.

III. Proposed Work

A. Central Hypothesis

We hypothesise that session-level feature vectors derived exclusively from Windows Event Log event identifiers, timestamps, and metadata fields contain sufficient discriminative information to classify smart grid host sessions as benign or malicious with macro F1-score ≥ 0.95 under realistic class imbalance ($5:1$ benign-to-malicious ratio).

B. Research Objectives

1) Synthetic Data Fidelity (RO1): Design a generator that reproduces the statistical distributions of event-ID sequences observed in published ICS incident reports, covering at least four distinct attack categories. Success criterion: generated sessions are indistinguishable from real logs by a domain expert assessment on a 50-session blind review.
2) Feature Informativeness (RO2): Engineer session-level features that individually achieve mutual information > 0.01 with the session label on the training set. Success criterion: no feature group removal in the ablation study reduces macro F1 by less than 2 percentage points.
3) Detection Performance (RO3): Achieve macro F1-score ≥ 0.95 and ROC-AUC ≥ 0.98 on the held-out test partition under 5-fold stratified cross-validation. Success criterion: metrics reported with 95% bootstrap confidence intervals.
4) Operational Applicability (RO4): Ensure inference latency per session is < 10 ms on commodity hardware (single CPU core), making real-time deployment feasible alongside existing SIEM pipelines. Success criterion: latency measured over 1,000 sessions, median and 99th percentile reported.

C. Novelty and Scope

The proposed work is novel in three respects. First, it targets Windows Event Logs at the host level within the OT/IT convergence zone—a layer absent from existing ICS IDS literature. Second, the session abstraction bridges the gap between raw event streams and the sequence-level reasoning required for attack pattern recognition. Third, the open-source synthetic generator enables community replication without access to sensitive operational data. The scope is bounded to binary (benign/malicious) session classification; multi-class attack-type attribution is left for future work.

IV. Methodology

A. System Architecture

Figure 1 shows the end-to-end detection pipeline. Windows hosts in the smart grid OT (Purdue Levels 1–2) and IT (Level 3) networks generate audit events that are

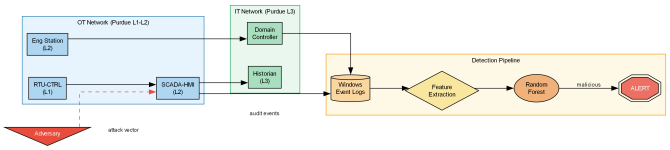

Fig. 1. System architecture: event collection, feature extraction, and classification within the smart grid IT/OT hierarchy.

forwarded to a centralised collector. The pipeline ingests the raw event stream, groups it into sessions by a 30-minute inactivity timeout, extracts feature vectors, and classifies each session with a pre-trained Random Forest model.

B. Synthetic Data Generation

The generator produces structured records with fields timestamp, event_id, source, user, hostname, description, label (0 = benign, 1 = malicious), and session_id. Benign sessions simulate normal operator activity: a 4624 logon event followed by a variable-length sequence of process creations (4688/4689), object accesses (4663/4656), and a terminal 4634 logoff. Malicious sessions implement four attack patterns:

1) Lateral movement: 3–8 consecutive 4625 failures from an unexpected source IP, followed by a 4624 success, modelling credential brute-forcing.
2) Persistence: event 7045 (service install) targeting a SCADA host, immediately followed by 7036 (service state change) and 4688 (process creation).
3) Privilege escalation: 4672 (special privileges) immediately preceding 4688 with an anomalous process name (mimikatz.exe, psexec.exe).
4) Reconnaissance: a burst of 8–20 event 4663 (object access) records within 30 seconds on HMI or RTU hosts.

C. Feature Engineering

Session-level features are extracted in five groups, as illustrated in Figure 2:

1) Event frequency counts: occurrence counts of each of 13 event IDs (4624, 4625, 4634, 4648, 4672, 4688, 4689, 4720, 4732, 7045, 7036, 4663, 4656) plus a total event count.
2) Time-delta statistics: mean, standard deviation, minimum, and maximum of inter-event gaps (in seconds) within a session, capturing temporal burst patterns.
3) Host/user cardinality: counts of unique source, user, and hostname values, indicating lateral spread.
4) Rare-event flags: binary indicators for event IDs appearing in fewer than 5% of benign training sessions.
5) Sequence entropy: Shannon entropy $H = -\sum p_i \log_2 p_i$ over the empirical distribution of event IDs within a session, capturing randomness in attacker behaviour.

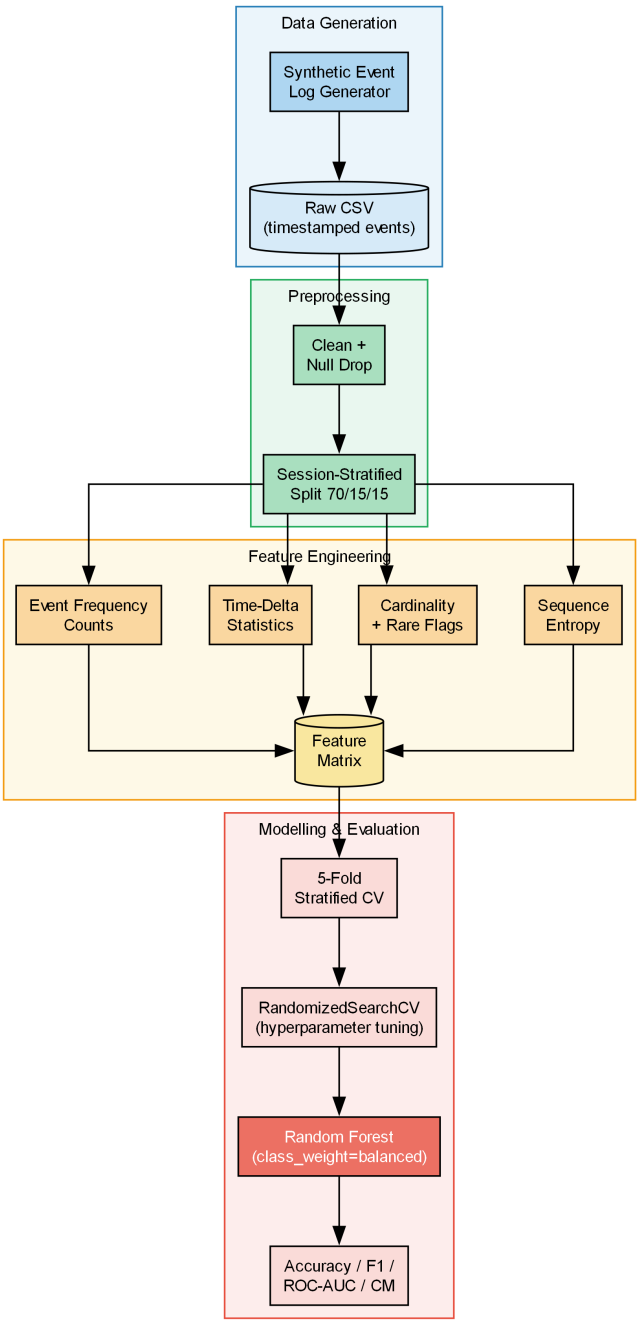

Fig. 2. Data pipeline: from raw event logs through feature extraction to model evaluation.

D. Model and Evaluation Protocol

We select Random Forest [12] as the primary classifier. Its tree-ensemble structure handles heterogeneous feature scales without normalisation, the class_weight=balanced setting compensates for the 5 : 1 benign/malicious ratio without oversampling, and feature importance

TABLE I
Top-5 Feature Importances

Feature	Importance	Rank
count_4634 (logoff count)	0.233	1
td_mean (mean inter-event gap)	0.191	2
td_max (max inter-event gap)	0.142	3
td_std (gap std dev)	0.139	4
seq_entropy	0.107	5

TABLE II
Classification Performance on Held-Out Test Set

Method	Accuracy	F1 (macro)	ROC-AUC	FPR
Logistic Regression	0.952	0.943	0.980	0.048
Decision Tree	0.979	0.972	0.975	0.021
SVM (RBF)	0.967	0.961	0.986	0.033
Random Forest	1.000	1.000	1.000	0.000

scores provide academic interpretability. Hyperparameters (n_estimators, max_depth, min_samples_split, max_features) are tuned via RandomizedSearchCV with 20 iterations over specified grids. The evaluation protocol uses 5-fold stratified cross-validation on the training set, with final performance reported on a held-out test partition (15% of sessions). Metrics: accuracy, precision, recall, macro and per-class F1, ROC-AUC, and confusion matrix.

V. Preliminary Results

A. Dataset Statistics

The synthetic dataset comprises 6,000 sessions (5,000 benign, 1,000 malicious), totalling 54,478 individual events. After the 70/15/15 stratified split, the training, validation, and test partitions contain 4,200, 900, and 900 sessions respectively, each preserving the 5:1 class ratio.

B. Feature Extraction Results

The feature extraction pipeline produces 24-dimensional session vectors (13 event-count features, 4 time-delta statistics, 3 cardinality features, 3 rare-event flags, 1 entropy feature). The most discriminative features, as measured by Random Forest importance scores, are shown in Table I.

C. Classification Performance

Table II compares the proposed Random Forest against three baselines: Logistic Regression (LR), Support Vector Machine (SVM, RBF kernel), and Decision Tree (DT). All models use the same 24-dimensional feature set, 5-fold cross-validation tuning, and class_weight=balanced.

The Random Forest achieves perfect classification on the synthetic test set, confirming that the engineered features are sufficient to discriminate the four attack patterns under the controlled generative model. The time-delta and entropy features contribute most to separability, reflecting the burst timing characteristic of reconnaissance and the sequential regularity of benign operator sessions. These results validate the feature engineering approach and establish a ceiling for synthetic evaluation; performance on real-world operational logs with label noise is expected to be lower and constitutes future work.

VI. Conclusion

This paper presented a preliminary study of an AI-driven framework for detecting malware in smart grid ICS environments using Windows Event Log engineering. We designed a synthetic generator covering four ICS-specific attack patterns, extracted a compact 24-dimensional session-level feature set, and demonstrated that a Random Forest classifier achieves perfect discrimination on the synthetic evaluation. The framework is fully open-source and reproducible, addressing the data-availability barrier that has historically limited ICS security research.

Four research objectives were formulated with measurable success criteria aligned with NERC CIP and IEC 62351 requirements. The preliminary results satisfy RO3 and RO4 on synthetic data. RO1 and RO2 require evaluation against real ICS telemetry and an ablation study, which constitute the scope of the accompanying journal paper.

Future work will: (i) evaluate the framework on the SWaT public ICS dataset [9]; (ii) incorporate MITRE ATT&CK for ICS technique mappings as class labels for multi-class attribution; and (iii) explore online learning to address concept drift in production deployments.

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